COMP 3270 – Homework 2
Algorithm Correctness and Loop invariant

1. Prove the correctness of the below given algorithm.

```
procedure CountingSort(arr):
    # Find the maximum and minimum values in the input
array
    max_val = findMax(arr)
    min_val = findMin(arr)

    # Create an array to store the count of each element in
the input array
    count = array of size (max_val - min_val + 1)
initialized with 0

    # Count the occurrences of each element in the input
array
    for num in arr:
        count[num - min_val] += 1

    # Update the count array to store the cumulative count
    for i from 1 to length(count) - 1:
        count[i] += count[i - 1]

    # Create the output array
    output = array of size length(arr)

    # Build the output array using the count array
    for num in reverse(arr):
        output[count[num - min_val] - 1] = num
        count[num - min_val] -= 1

    return output
```

2. Consider the following pseudo code, which implements the selection sort. It works by looking through an array of n elements, A, picking the smallest element and moving it to the 1st position. It then, repeats the same thing on the sub-array, A[2..n], defined by positions 2...n, then, A[3..n], defined on positions 3..n and so on.

```
1     i := 1
2     while i <= n do {
3         small := i
4         j = i + 1
5         while j <= n do {
6             if A[j] < A[small] then small := j fi
7             j := j + 1
8             }
9     temp := A[small]
10    A[small] := A[i]
11    A[i] := temp
12    i := i + 1
13    }
```

A. What is the best loop invariant for the inner loop (lines 5..8)?
B. What is the best loop invariant for the outer loop (lines 2-13)?
C. Use the inner loop invariant to show the correctness of the inner loop (lines 3..8).

3. For the algorithm given below,

```
Input: an array, 'arr' of length n
Output: reversed array, 'arr'
def reverse_array(arr):
    left, right = 0, len(arr) - 1
    while left < right:
        arr[left], arr[right] = arr[right], arr[left]
        left += 1
        right -= 1
```

a. State the loop invariant.
b. Use the invariant to prove that the postcondition is satisfied.
c. Show that the algorithm terminates.

4. Prove that sum = i^2 after every iteration of the for loop below:

Input: n: nonnegative integer
Output: n^2

$sum = 0$
for $i = 1$ to n **do**
  $sum = sum + 2i - 1$
end
return sum